

Tie-out roster: the whole credit suite

Every data point in both monitors, checked against the body that produces it. 5 September 2026.

776 of 778 data points tie. 2 differ. 0 could not.

Every figure on the ours side was read out of the shipped workbook — the cell a person opens — and never re-fetched. Every figure on the other side came off a document published by somebody else: a bank's own filed Call Report, or the agency that computes a series. The two that differ are set out first.

What does not tie

Two lines, both PNC Bank, and the workbook is not the side that is wrong.

PNC's credit card and C&I net charge-offs for the quarter differ from the bank's own filed report by 515 and 652 thousand dollars, about half a percent. The workbook carries the FDIC's published quarterly figure to the dollar, and the FDIC's own year-to-date figure matches PNC's filing exactly. What does not reconcile is the FDIC's own arithmetic:

FDIC quarterly	Q1	43,842	+	Q2	47,617	=	91,459	
FDIC year-to-date at Q2						=	91,974	off by 515
PNC's filed Call Report						=	91,974	
FDIC quarterly	Q1	101,857	+	Q2	109,201	=	211,058	
FDIC year-to-date at Q2						=	211,710	off by 652
PNC's filed Call Report						=	211,710	

The same reconciliation across all twelve banks and all three quarterly flow fields holds 34 times out of 36. Only these two fail. No merger explains it: PNC's only 2026 acquisition events are branch transfers dated 6 July 2026, after the reporting date.

I believe the filing, because it is the document the bank signed and the FDIC's own annual figure agrees with it. Nothing was adjusted. The workbook faithfully carries what its source published, and this is written down rather than plugged.

What the tie-out found

Six defects, and every one of them left the numbers correct, which is why a test suite of 414 passing tests had never seen any of them.

- 1. **A shipped workbook with a state missing from it.** Nebraska's house price index was blank. One `Internal Server Error` from the data provider blanked the slot and the build passed. The retry covered rate limits only. *Fixed.*
- 2. **Two series wearing each other's description** — the commercial real estate construction and nonfarm-nonresidential labels were swapped on the dashboard and in the raw tabs. *Fixed.*
- 3. **A tightening indicator with its alert switched off**, because it had been filed as a demand series, and demand series correctly get no alert. A real mortgage-tightening signal could never fire. *Fixed, and that turns an alert on.*
- 4. **Two more labels naming a different series** than the number beside them. *Fixed.*
- 5. **Four series declaring “billions” beside a figure in millions** — a factor of a thousand on the line a person reads. *Fixed.*
- 6. **The FDIC's own quarterly and annual figures for PNC do not reconcile.** Reported above; nothing changed, because our side is right.

And three defects in the checking itself, which is the likelier culprit than the code and was here too. My first pass cited the wrong Call Report line for C&I charge-offs (U.S. addressees only, where the measure is U.S. and non-U.S.), took the wrong column of the total capital ratio for the one bank that files two, and produced six blank source photographs that reported “ok”. Each showed up as an implausibly uniform failure across every entity — which is the tell.

The denominator, by set					
Set	Checked against	Lines	Tie	Differ	
JPMorgan Chase Bank NA cert 628	its own filed Call Report, 30 June 2026	53	53	0	
Wells Fargo Bank NA cert 3511	its own filed Call Report, 30 June 2026	53	53	0	
Bank of America NA cert 3510	its own filed Call Report, 30 June 2026	53	53	0	
Citibank NA cert 7213	its own filed Call Report, 30 June 2026	53	53	0	
US Bank NA cert 6548	its own filed Call Report, 30 June 2026	53	53	0	
PNC Bank NA cert 6384	its own filed Call Report, 30 June 2026	53	51	2	

Set	Checked against	Lines	Tie	Differ
Truist Bank cert 9846	its own filed Call Report, 30 June 2026	53	53	0
Capital One NA cert 4297	its own filed Call Report, 30 June 2026	53	53	0
KeyBank NA cert 17534	its own filed Call Report, 30 June 2026	53	53	0
Bank of New York Mellon cert 639	its own filed Call Report, 30 June 2026	53	53	0
Goldman Sachs Bank USA cert 33124	its own filed Call Report, 30 June 2026	53	53	0
Morgan Stanley Bank NA cert 32992	its own filed Call Report, 30 June 2026	53	53	0
FHFA All-Transactions house price indexes	Federal Housing Finance Agency	67	67	0
Charge-off and delinquency rates at commercial banks	Federal Reserve Board	29	29	0
G.19 consumer credit, the debt service ratios, FHFA monthly	Federal Reserve Board and FHFA	9	9	0
Senior Loan Officer Opinion Survey	Federal Reserve Board	13	13	0
S&P Cotality Case-Shiller house price indexes	S&P Dow Jones Indices	22	22	0
Z.1 commercial property price indexes	Federal Reserve Board	2	2	0
Total		778	776	2

Where the evidence is

One document per bank and one for the whole macro set. Each is self-contained: every screenshot is embedded, so it survives being forwarded to somebody who does not have this machine.

Document	What is in it
banks-12-2026-06-30/TIE-OUT-Bank-of-America-NA-cert3510-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it

Document	What is in it
banks-12-2026-06-30/TIE-OUT-Bank-of-New-York-Mellon-cert639-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Capital-One-NA-cert4297-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Citibank-NA-cert7213-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Goldman-Sachs-Bank-USA-cert33124-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-JPMorgan-Chase-Bank-NA-cert628-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-KeyBank-NA-cert17534-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call

Document	What is in it
	Report that carries it
banks-12-2026-06-30/TIE-OUT-Morgan-Stanley-Bank-NA-cert32992-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-PNC-Bank-NA-cert6384-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Truist-Bank-cert9846-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-US-Bank-NA-cert6548-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Wells-Fargo-Bank-NA-cert3511-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
fred-142-series-2026-09-05/TIE-OUT-fred-142-series-2026-09-05.pdf	142 series across six publishers, with every workbook cell

Document	What is in it
	photographed and each agency's own page marked
keybank-card-30-89-2026-09-05/	the first exhibit: one figure traced link by link, and all 53 KeyBank lines

What this does not prove

- **One quarter for the banks, one observation for the macro series.** The bank workbook holds sixteen quarters each and the macro workbook holds 13,905 observations. What was compared is the most recent of each — 778 figures, not the whole history. The Case-Shiller set was additionally checked a second month.
- **Raw inputs, not the ratios built on them.** Forty-eight of each bank's lines are dollar amounts read straight off the form, and two are ratios the bank files itself. The FDIC's computed ratios, and the dashboard's z-scores, bands and alert logic, are arithmetic over figures this proves — and are not themselves checked here. That was the instruction: the base data.
- **Twenty of the Case-Shiller series are tied through a ratio of two cells,** not a level, because S&P does not publish adjusted levels free. That pins the month-on-month move, not the absolute level.
- **The provenance map passed rather than was proved universal.** Every code it cites was found on every one of these twelve filings and every number matched. A citation that is right for these filers and wrong for a bank filing a different form would not show up here.
- **Nothing here was checked by a second person.**